

Python Machine Learning (CSCI 425) Student Showcase (Final Project) Rubric

Project Title: **INTRUSION DETECTION WITH MACHINE LEARNING**
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Instructions:

- For each requirement, use 0.0 - 5.0 scale in the Score column (0 - Fail, 1 - Needs improvement, 2 - Poor, 3 - Fair, 4 - Good, 5 is Excellent)

Criteria	Score (1-5)
1. Problem Definition & Motivation	5
a. Clearly defined problem statement.	5
b. Understanding of the domain and context	5
2. Data Acquisition & Preprocessing	5
a. Data cleaning and handling missing values	5
b. Feature engineering and selection	5
c. Data normalization and scaling	5
3. Model Selection, Evaluation & Justification	5
a. Selection of appropriate ML algorithms	4
b. Model training and tuning	4
c. Evaluation metrics and performance analysis	5
4. Creativity and Innovation	5
a. Novelty and originality of approach	4
b. Exploration of advanced techniques (Deep Learning, e.g.)	4
5. Presentation	5
a. Quality of visualizations and insights	5
b. Ability to communicate results effectively	4
Final Score	55/60

XGBoost
→ but no
NN

General Feedback:

Some ~~features~~ features would be unavailable until the session ends, like "session length".

Login reputation score (new feature) would ~~become~~ ^{scale} interestingly based on the multiplication

I think some of the graphs (ROC curves) may require some tweaking (No false positives)

Did you try to isolate performance on the login reputation score?